



Building_Energy_Modeling_Secondary_School_Layout_Investigation

Secondary School Layout Investigation Report

Overview

This report explains why the full-size secondary school does not fit within its allocated ground in one neighbourhood layout, and compares it with a case where a school fits easily on a 70m x 70m plot.

1. Dimensional Analysis

Current Building

The simulation uses the **ASHRAE 90.1 Secondary School (Buffalo)** prototype model.

- **Footprint Area:** ~11,902 m²
- **Actual Dimensions:** 141.0m (Width) x 104.0m (Depth)
- **Building Height:** 8.0m (2 floors)

Allocated Plot

The secondary school is assigned to a plot within the neighbourhood layout.

- **Plot Dimensions:** 142.5m x 71.2m
- **The Conflict:** While the building's width (141m) just barely fits the plot width (142.5m), the **104m depth significantly exceeds the 71.2m plot depth**. This causes the building to overflow the ground by **32.8 meters** in the depth direction.

2. Comparison with the 70m x 70m Example

A comparison example shows a school fitting into a 70m x 70m ground with room to spare. This is possible because it uses a **resized** version of the school.

Resized School Model

The project also contains a resized version of the school, developed for a different site:

- **Footprint Area:** ~1,266 m² (nearly 10 times smaller than the Buffalo version)
- **Approximate Dimensions:** ~45m x ~33m
- **Fit:** This resized version fits easily within a 70m x 70m (4,900 m²) plot and even leaves enough space for additional buildings on the same ground.

3. Root Cause

The neighbourhood layout is hardcoded to use the full-sized Buffalo school rather than the resized model, causing the depth overflow described above.

4. Conclusion & Recommendation

The secondary school does not fit because the **Buffalo Secondary School model is physically too deep (104m) for the designated ground (71.2m)**.

To resolve the layout issue and match the efficiency seen in the comparison example, the neighbourhood configuration should be updated to use a resized school model (like the Calgary resized version) that aligns with the intended site density.